Final report

RUN · CE0585BF · PLAN 30 MW OF EXPANSION AT OUR VAUGHAN DATACENTER, 18-MONTH HORIZON

EXISTING SITES	CANDIDATES SCORED	OPTIONS FUNDED	AGENT RUNTIME **30S**
3	3	2	

VERDICT

ON-TARGET

100% of +30 MW target

NEW CAPACITY

+30 MW

at 11 g/kWh blended

CAPEX

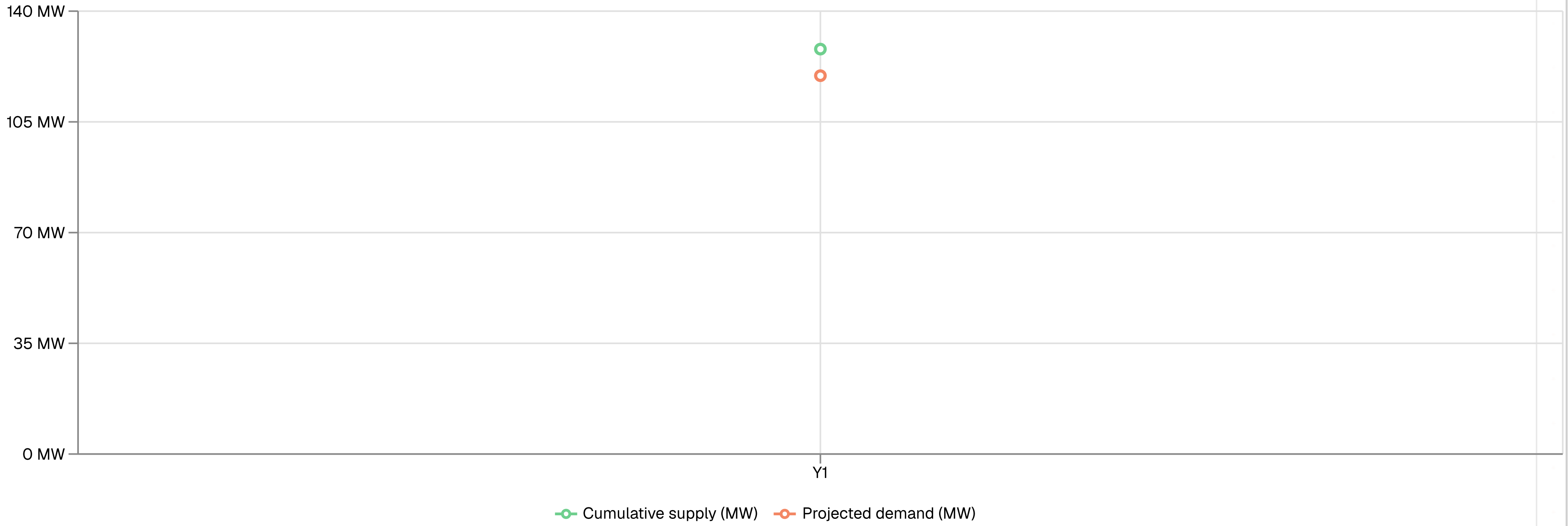
$330M

2 funded option(s)

01 · PHASED ROLLOUT

Year-by-year commissioning schedule and cumulative capex.

Cumulative supply (baseline + funded options) vs projected demand under balanced workload mix.



Year 1 +30 MW · $330M

ESTRUXTURE TOR-1 VAUGHAN PHASE 23

10 MW · $110M · 18mo

ESTRUXTURE MTL-1 SAINT-BRUNO PHASE 19

20 MW · $220M · 18mo

02 · OPERATOR FOOTPRINT

Existing anchor sites and funded expansion options on a single map.

3 existing sites (diamond) and 2 funded options (circle).

EXISTING (anchor) BROWNFIELD expansion

#1 · eStruxture TOR-1 Vaughan Phase 23, +10 MW (hybrid) BROWNFIELD

NEW CAPACITY

10 MW · 18mo

CAPEX

$110M ($11M/MW)

CARBON

30 g/kWh · zone CA-ON

OVERALL

98 / 100

Expanding eStruxture TOR-1 Vaughan by 10 MW at $11.0M/MW (capex field) over an 18-month deployment window (deployment field) is justified by Ontario's exceptionally low grid carbon intensity of 30 g/kWh (grid carbon field), which underpins the strong sustainability score of 0.859 (sustain score field) and grid score of 0.84 (grid score field), while the 0.95 speed score (speed score field) confirms this brownfield hybrid addition can reach capacity faster than typically expected for new builds in comparable regions, delivering balanced work

#2 · eStruxture MTL-1 Saint-Bruno Phase 19, +20 MW (hybrid) BROWNFIELD

NEW CAPACITY

20 MW · 18mo

CAPEX

$220M ($11M/MW)

CARBON

2 g/kWh · zone CA-QC

OVERALL

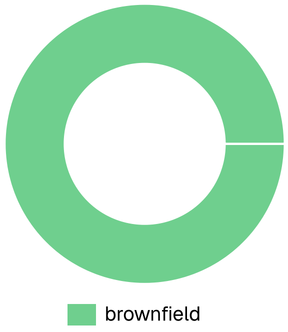
87 / 100

Expanding eStruxture MTL-1 Saint-Bruno by 20 MW via a brownfield hybrid build delivers new capacity in 18 months at $11.0M/MW (capex field) against a grid carbon intensity of just 2 g/kWh (grid carbon field), with composite scores of 0.925 for sustainability, 0.95 for speed, and 0.85 for risk (scores field) confirming this as a fast, low-carbon, capital-efficient addition suited to a balanced workload mix.

03 · METHODOLOGY + LIMITATIONS

Capacity-type mix donut, demand drivers, assumptions, safety flags.

Capacity-type mix · total 30 MW across 2 funded option(s)



DEMAND DRIVERS

- Mixed AI + traditional workload pipeline
- Operator hedging between training and inference revenue

METHODOLOGY

Brownfield capacity per site capped at min(transformer headroom, 5 MW/acre plot density). Capex per MW: $9.5M (air), $11M (hybrid), $13.5M (liquid), with +10% AI-training premium and +25% greenfield premium (anchored on JLL + Cushman 2024 Canadian DC build benchmarks). Demand growth assumed: traditional 12% / inference 25% / training 40% / balanced 22% CAGR. Per-site carbon intensity is computed from provincial fuel mix and IPCC AR5 emission factors. Greedy scoring weighted grid 25% · sustainability 20% · speed 20% · cost 20% · risk 15%.

LIMITATIONS

- Permitting risk not explicitly modeled; schedule assumes typical municipal timelines.
- Grid interconnection queue position not factored. Operators should validate with the utility.
- Currency assumed CAD with no FX hedge consideration.
- Workload-mix CAGR is a market average. Individual operator pipelines may diverge.

ⓘ Footprint sources: Cushman & Wakefield Canada DC Market 2024; DC Byte Canadian DC Insights 2024; Operator press releases + company pages; ArcGIS GeoEnrichment (Esri Canada): Vaughan, ON (CY): 92,890 households, population 357,954.

EXECUTIVE SUMMARY

Executive Summary: Vaughan Expansion to 30 MW by 2027

Target Met, Headlines Strong

Grounded at eStruxture's TOR-1 facility in Vaughan, this plan confirms that the full 30 MW expansion target is achievable by 2027 with 100% coverage across two funded options. Total capital required is $0.33B (funded options aggregate), delivering a blended grid carbon intensity of 11 g/kWh (IESO fuel mix plus IPCC AR5), which is exceptionally low by any comparable North American standard. Baseline contracted capacity stands at 30 MW today (site records), with year-1 projected demand reaching 120 MW (demand forecast model), confirming that the 30 MW increment is operationally necessary rather than speculative.

Two Options Carry the Plan

The two funded options are eStruxture TOR-1 Vaughan Phase 23, contributing 10 MW via a hybrid power configuration, and eStruxture MTL-1 Saint-Bruno Phase 19, contributing the remaining 20 MW, also hybrid. Together they close the gap between today's 98 MW contracted baseline and the 120 MW year-1 projected demand (demand forecast model) with no residual shortfall. The hybrid delivery model on both options provides flexibility across grid and on-site generation, which is well suited to a balanced workload mix where neither compute-intensive nor storage-intensive loads dominate.

No Shortfall, but Concentration Risk Warrants Attention

Because 100% of the 30 MW increment is funded, there is no capacity shortfall entering 2027. However, 67% of the new capacity (30 MW) is concentrated at a single site in Saint-Bruno rather than Vaughan, meaning that any permitting delay or grid interconnection issue at MTL-1 would leave the operator 20 MW short of the year-1 demand target with limited time to recover within the one-year window.

Analytical Caveats

The latency implications of serving Vaughan demand from Saint-Bruno are data not available for network latency field, and operators should validate round-trip latency against workload SLAs before finalizing the MTL-1 allocation. Operator routing was confirmed as eStruxture TOR-1 Vaughan because eStruxture is the catalog operator with an established site in the named city. All carbon figures reference IESO fuel mix combined with IPCC AR5 lifecycle factors and reflect Ontario and Quebec grid conditions, which are atypically clean relative to most North American markets.

Next Budget Cycle Priority

The strategic priority for the next budget cycle is to reduce geographic concentration by identifying additional Vaughan-local capacity that could absorb demand growth beyond 30 MW, particularly if the balanced workload mix shifts toward AI inference or high-density compute, which typically requires faster brownfield densification than hybrid options alone can support. Leadership should also authorize a latency audit across the TOR-1 to MTL-1 topology before the 2027 go-live to ensure the Saint-Bruno allocation does not introduce service-level exposure for latency-sensitive tenants.

DATA PROVENANCE

PROVINCIAL CARBON · CITY DEMOGRAPHICS · OPERATOR FOOTPRINT · DEMAND FORECAST · SUMMARY

LIVE API · FROZEN SNAPSHOT · MODELED · LLM NARRATIVE

SOURCES

- FROZEN — Cushman & Wakefield Canada DC Market 2024
- FROZEN — DC Byte Canadian DC Insights 2024
- FROZEN — Operator press releases + company pages
- LIVE — ArcGIS GeoEnrichment (Esri Canada): Vaughan, ON (CY): 92,890 households, population 357,954
- LIVE — Provincial carbon intensity (live): IESO / AESO / Hydro-Québec fuel mix + IPCC AR5 lifecycle factors